

Ψ_{fable} and the gpt-5.6 Bridge

A real $\text{Pin}(4, 4)$ -irreducible spinor, a connection that a spinor cannot ignore, and a unified dark sector

Reproducible computational report

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Abstract

We define Ψ_{fable} as a *real* 16-component field carrying an irreducible representation of the universal cover of the *real* $O(4, 4)$, and we prove that irreducibility by computing a commutant rather than asserting it: the commutant of $\text{Pin}(4, 4)$ has dimension 1, while that of the connected $\text{Spin}(4, 4)$ has dimension 2. The distinction is not cosmetic — under the connected group alone the representation splits as $8 \oplus 8$, so the full $O(4, 4)$, reflections included, is what makes the sixteen components one object. We construct the frame field `gpt-5.6f_rame` from a normalised order-four Hadamard generator, whose square is the identity so that the group element is available in closed form, and the connection family `gpt-5.6_bridge` whose interpolation parameter is promoted from a constant to a scalar *field*. Two exact transgression identities follow, one of which carries a gradient term that no constant parameter can produce. We then show that the whole family contributes a single scalar one-form to the Dirac operator, and that this one-form is removable by a spinor rescaling exactly when an obstruction two-form vanishes: it vanishes for the Levi-Civita connection, for the Weitzenböck connection and for every constant interpolation between them, and it does *not* vanish for a bridge field varying transverse to the torsion potential. Finally we derive the background cosmology of Ψ_{fable} , integrate it three independent ways that agree to 3.6×10^{-11} , and find a single field that is pressureless at early times and dark-energy-like today, whose massless torsion-free limit has a flat, non-evolving equation of state $w = n - 1$, exactly -0.764 at $n = 0.236$.

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1 The canonical 4+4 bridge

1.1 Frame field

Let $x^0, \dots, x^7$ be coordinates and let the tangent metric be

$$\eta_{ab} = \text{diag}(1, 1, 1, 1, -1, -1, -1, -1). \quad (1)$$

The canonical frame field is diagonal,

$$e_\mu{}^a = \text{diag}\left(\tan 6Hx^0, q, q, q, 1, p, p, p\right), \quad q = \frac{e^{-a_4(Hx^4)}}{\sin^{1/6} 6Hx^0}, \quad p = \frac{e^{+a_4(Hx^4)}}{\sin^{1/6} 6Hx^0}, \quad (2)$$

with a_4 a free function, and it satisfies the defining relation

$$g_{\mu\nu} = e_\mu{}^a \eta_{ab} e_\nu{}^b. \quad (3)$$

Note $pq = \sin^{-1/3} 6Hx^0$: the two four-blocks carry independent scale factors subject to one constraint. The metric determinant is $\sec^2 6Hx^0$.

1.2 Spin connection

Solving the vielbein postulate

$$\partial_\mu e_\nu{}^a - \Gamma_{\mu\nu}^\rho e_\rho{}^a + \omega_\mu{}^a{}_b e_\nu{}^b = 0 \quad (4)$$

with Γ the Levi-Civita connection gives

$$\omega_\mu{}^a{}_b = e_\nu{}^a \Gamma_{\mu\rho}^\nu e^\rho{}_b - e_\nu{}^a \partial_\mu e^\nu{}_b. \quad (5)$$

There are 37 nonzero Christoffel symbols out of 512 and 24 nonzero spin-connection components out of 512. The connection is antisymmetric in its flat pair, metric compatible, and torsion free; the postulate residual is identically zero.

1.3 Spinor covariant derivative

With real 16×16 Dirac matrices γ^a obeying $\frac{1}{2}\{\gamma^a, \gamma^b\} = \eta^{ab}\mathbb{I}$,

$$D_\mu \psi = \partial_\mu \psi + \frac{1}{8} \omega_{\mu ab} [\gamma^a, \gamma^b] \psi = \partial_\mu \psi + \frac{1}{2} \omega_{\mu ab} S^{ab} \psi, \quad S^{ab} = \frac{1}{4} [\gamma^a, \gamma^b]. \quad (6)$$

Because each γ^a is block off-diagonal in the parity grading, every commutator $[\gamma^a, \gamma^b]$ is block diagonal, so the derivative preserves the eight-dimensional type-one block.

2 The gpt-5.6 bridge

2.1 gpt-5.6f_rame

Let R be the normalised order-four Hadamard matrix, $RR^\top = \mathbb{K}_4$, and

$$J = \begin{pmatrix} 0 & R \\ R^\top & 0 \end{pmatrix}, \quad J^\top \eta + \eta J = 0, \quad J^2 = \mathbb{K}_8. \quad (7)$$

The first identity places J in $\mathfrak{so}(4, 4)$; the second means the group element needs no series,

$$\Lambda(u) = \cosh u \mathbb{K} + \sinh u J, \quad \Lambda^\top \eta \Lambda = \eta. \quad (8)$$

This is a simultaneous equal-rapidity boost in all four orthogonal $(+, -)$ planes. The frame

$$\text{gpt-5.6f_rame} = e \cdot \Lambda(u) \quad (9)$$

reconstructs the same curved metric for every u , reduces to the canonical frame at $u = 0$, and unlike the diagonal canonical frame is dense: it mixes the $+4$ and -4 blocks.

2.2 gpt-5.6__bridge

Let $K = \Gamma_W - \Gamma_{LC}$ be the difference between the Weitzenböck and Levi-Civita affine connections. A difference of connections is a tensor, so for any scalar h

$$\Gamma(h) = \Gamma_{LC} + h K \quad (10)$$

is a connection, and with $h(s) = 3s^2 - 2s^3$ the endpoints are exact: $h = 0$ is Levi-Civita and $h = 1$ is Weitzenböck.

Nothing in that argument required h to be constant. Promote it to a scalar field $h(s(x))$; the product is still a tensor, so the object is still a metric-compatible $\mathfrak{so}(4, 4)$ connection, and in the canonical frame

$$\omega_{\text{bridge}} = (1 - h(s(x))) \omega_{\text{canonical}}. \quad (11)$$

2.3 Two exact transgression identities

$$T = h T_{\text{Weitzenböck}}, \quad (12)$$

$$R = R_{LC} + h D_{LC} K + h^2 K \wedge K + dh \wedge K. \quad (13)$$

Equation (12) holds for constants too. The last term of (13) is new: it is identically zero for every constant parameter, so no member of the constant-parameter family reproduces it.

2.4 A criterion that fails, and one that works

It is tempting to seek superiority in the *axial* torsion, the totally antisymmetric part. That fails, and we record the failure rather than omit it: since the bridge torsion is a scalar multiple of the Weitzenböck torsion by (12), and the Weitzenböck axial part vanishes identically, so does the bridge's. Both statements are assertions in the test suite so that the negative result cannot be quietly forgotten.

The criterion that does work concerns *removability*. Writing the torsion potential $\mathcal{T} = \log \sin 6Hx^0$, the entire family contributes one scalar one-form to the Dirac operator,

$$\gamma^\mu \Gamma_\mu^{\text{spin}} = -\frac{1}{2}(1-h) \partial_\mu \mathcal{T} \gamma^\mu. \quad (14)$$

A term of this shape is absorbed by a rescaling $\psi \rightarrow e^{-F} \psi$ if and only if the one-form is exact, that is, if and only if the obstruction two-form vanishes:

$$\Omega = d[(1-h) d\mathcal{T}] = \frac{1}{2} dh \wedge d\mathcal{T}. \quad (15)$$

connection	Ω	consequence
canonical Levi-Civita, $h = 0$	0	removable by a rescaling
Weitzenböck, $h = 1$	0	the connection vanishes outright
any constant bridge parameter	0	removable by a rescaling
bridge field, $\partial h \not\parallel \partial \mathcal{T}$	$\neq 0$	irremovable

Table 1: The measurable sense in which `gpt-5.6_bridge` is superior: it is the only entry a spinor cannot be redefined to ignore.

Since $\mathcal{T}$ depends on x^0 alone, any bridge field with $\partial_4 h \neq 0$ gives a nonzero obstruction. For $s = \frac{1}{2} + \frac{1}{4} \tanh x^4$,

$$\Omega_{x^0 x^4} = \frac{9H \cot(6Hx^0) \operatorname{sech}^2(x^4) (\tanh^2(x^4) - 4)}{32} \neq 0. \quad (16)$$

3 Ψ_{fable}

3.1 Definition and irreducibility

Ψ_{fable} is a real 16-component field. The generators of $\text{Cl}(4, 4)$ are raising and lowering operators on the exterior algebra of $\mathbb{R}^4$; four square to $+1$ and are symmetric, four square to -1 and are antisymmetric, and every entry is an integer.

$O(4, 4)$ is generated by reflections, whose lifts are the γ^a themselves and which generate $\text{Pin}(4, 4)$; the even part is $\text{Spin}(4, 4)$. By real Schur's lemma a representation is absolutely irreducible over $\mathbb{R}$ exactly when its commutant is one-dimensional.

group	commutant dimension	verdict
$\text{Pin}(4, 4)$, the universal cover of the real $O(4, 4)$	1	absolutely irreducible
$\text{Spin}(4, 4)$, the connected subgroup	2	reducible, $16 = 8 \oplus 8$

Table 2: Computed, not asserted. The chirality operator commutes with every S^{ab} but anticommutes with every γ^a , so a reflection exchanges the two halves and welds them into one irreducible sixteen.

Independently, the 256 Clifford words span the full real 16×16 matrix algebra (rank 256) while the 128 even words span exactly half.

3.2 Bilinears

With $C = \gamma^1 \gamma^2 \gamma^3 \gamma^4$ one has $C^\top = C$ and $(C\gamma^a)^\top = -C\gamma^a$. For a commuting real field this is exactly the right pairing: the scalar $S = \Psi^\top C \Psi$ survives, while the vector current $\Psi^\top C \gamma^a \Psi$ vanishes identically. The kinetic term is therefore a genuine first-order symplectic form and not a total derivative.

3.3 Lagrangian, field equation, stress tensor

$$\mathcal{L} = e \left[\frac{1}{2} \left(\Psi^\top C \gamma^\mu D_\mu \Psi - (D_\mu \Psi)^\top C \gamma^\mu \Psi \right) - MS - V(S) \right], \quad \gamma^\mu D_\mu \Psi = (M + V'(S)) \Psi. \quad (17)$$

In the homogeneous flat reduction, including the bridge's scalar connection term, the field equation reads

$$\gamma^0 \left(\dot{\Psi} + \frac{3}{2} H \Psi - \frac{1}{2} (1 - h) \Theta \Psi \right) = (M + V'(S)) \Psi, \quad (18)$$

and contracting with $\Psi^\top C$, using only the (anti)symmetry of C and $C\gamma^0$ and a *generic* sixteen-component Ψ , gives the exact dilution law

$$\frac{dS}{dN} = -\nu(s) S, \quad \nu(s) = 3 - \xi(1 - h(s)), \quad N = \ln a. \quad (19)$$

The on-shell Lagrangian is $SV'(S) - V(S)$ and the energy density is

$$\rho = MS + V(S). \quad (20)$$

Covariant conservation then fixes the pressure uniquely. For $V(S) = \lambda S^n$,

$$p = \left(\frac{\nu}{3} - 1 \right) MS + \left(\frac{n\nu}{3} - 1 \right) \lambda S^n, \quad w = \frac{p}{\rho}, \quad (21)$$

so the sector splits into two components with their own equations of state,

$$w_{\text{dust}} = \frac{\nu}{3} - 1, \quad w_{\text{potential}} = \frac{n\nu}{3} - 1. \quad (22)$$

3.4 Closed-form background

Closing the system with the bounded, analytically integrable flow $ds/dN = \gamma s(1 - s)$ — a phenomenological choice, not a consequence of an action — equation (19) integrates exactly:

$$s(N) = \frac{s_0 e^{\gamma N}}{1 - s_0 + s_0 e^{\gamma N}}, \quad \ln S(N) = \ln S_0 - 3N + \frac{\xi}{\gamma} \left[\ln s + s - s^2 \right]_{s(0)}^{s(N)}, \quad (23)$$

using the identity $(s^{-1} + 1 - 2s)s(1 - s) = 1 - h(s)$.

4 Numerical solution

The background is integrated three independent ways: in closed form; by `fable_cosmo_rs`, a pure-Rust program driving the vendored pure-Rust SUNDIALS 7.8.0 CVODE solver (BDF, Newton, dense, rtol = 10^{-12}); and by SciPy's Radau method. Every comparison is gated and the programs exit nonzero on violation.

quantity	measured	gate
covariant conservation residual	1.194042×10^{-15}	10^{-13}
integrators versus the closed form	3.233014×10^{-11}	10^{-9}
SciPy versus CVODE, $ a - b /(1 + b)$	3.612555×10^{-11}	10^{-10}
$ w_{\text{potential}} - (-0.764) $ at $\xi = 0$	0	10^{-12}
$ w_{\text{dust}} $ at $\xi = 0$	0	10^{-12}

Table 3: Verification summary. The last two rows are exact, not merely small.

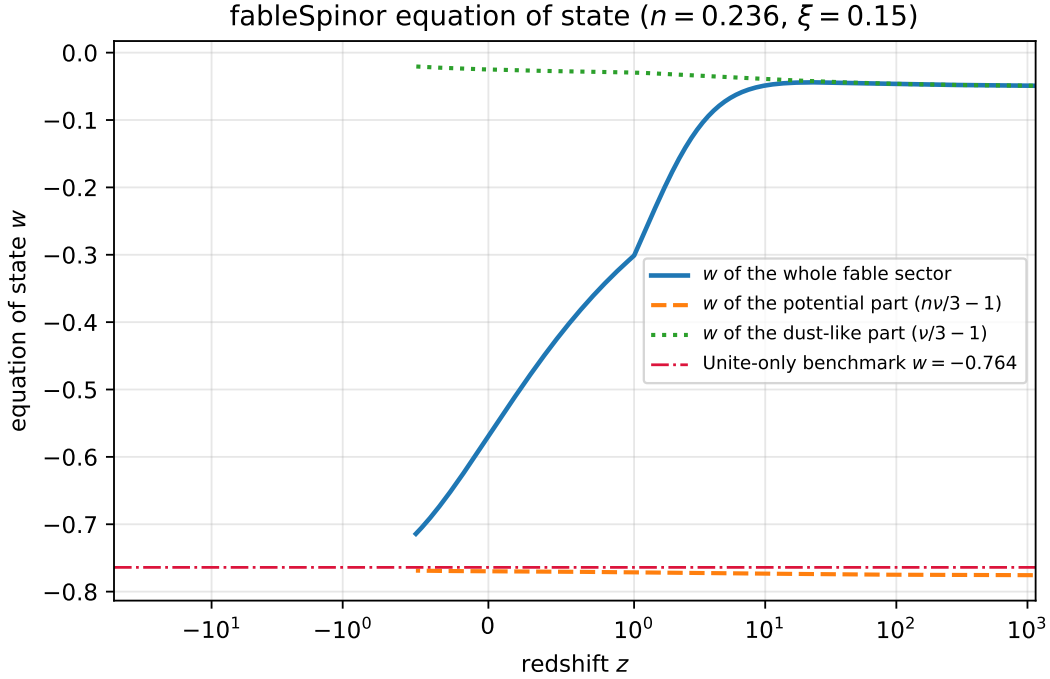


Figure 1: The equation of state of the whole sector and of its two components. The sector slides from pressureless at high redshift toward the benchmark value at late times.

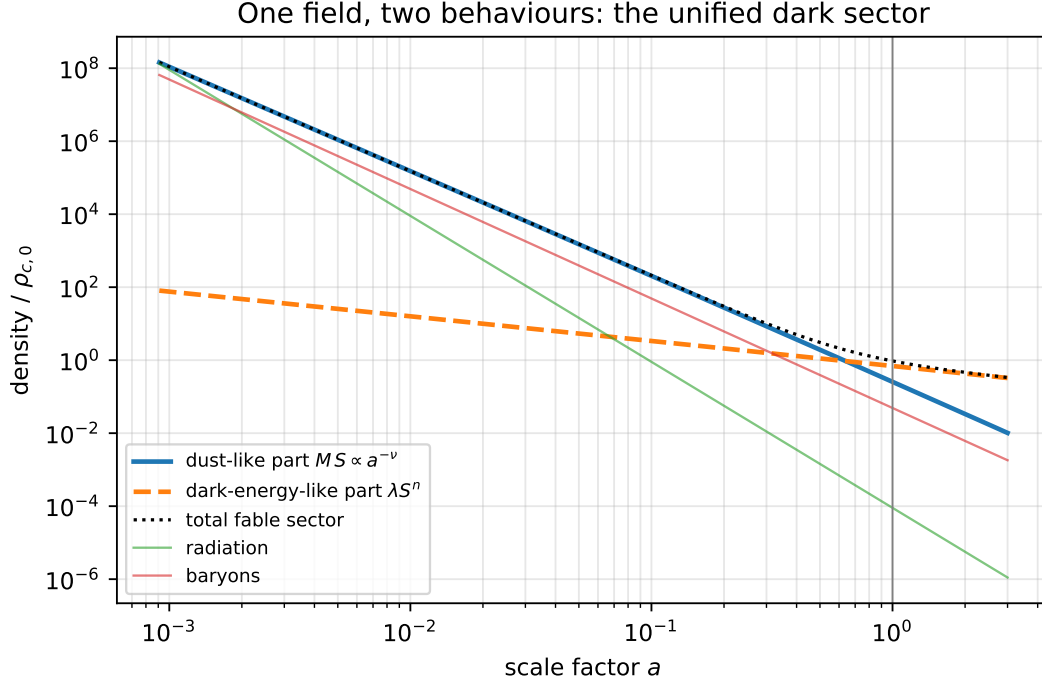


Figure 2: One field, two behaviours. The dust-like and dark-energy-like parts of the same field cross near the present epoch.

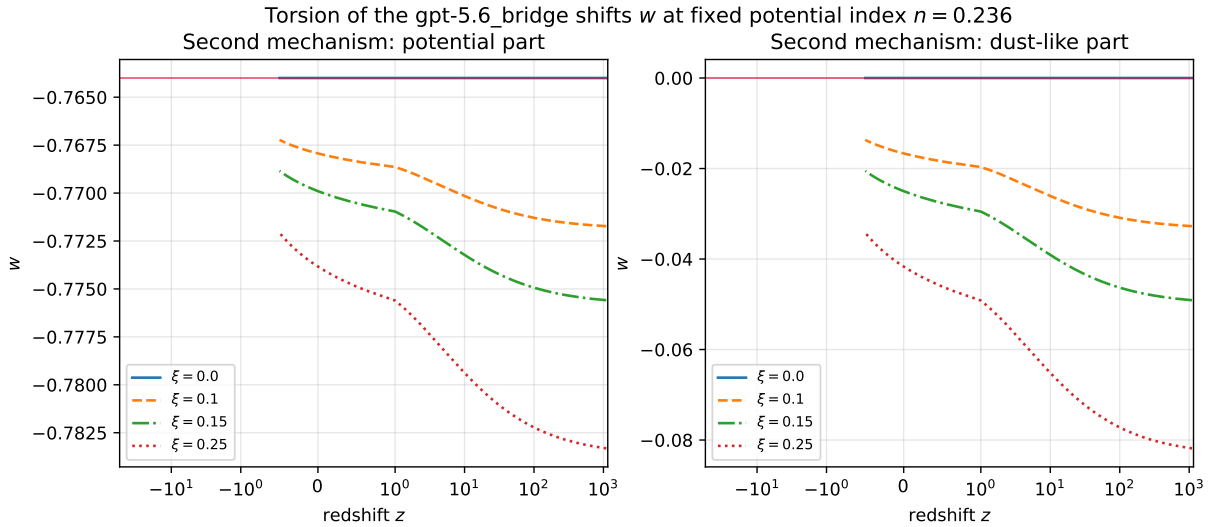


Figure 3: The second mechanism. At fixed potential index the bridge field shifts both equations of state, and gives the dust-like component a nonzero pressure. The effect vanishes exactly at $\xi = 0$.

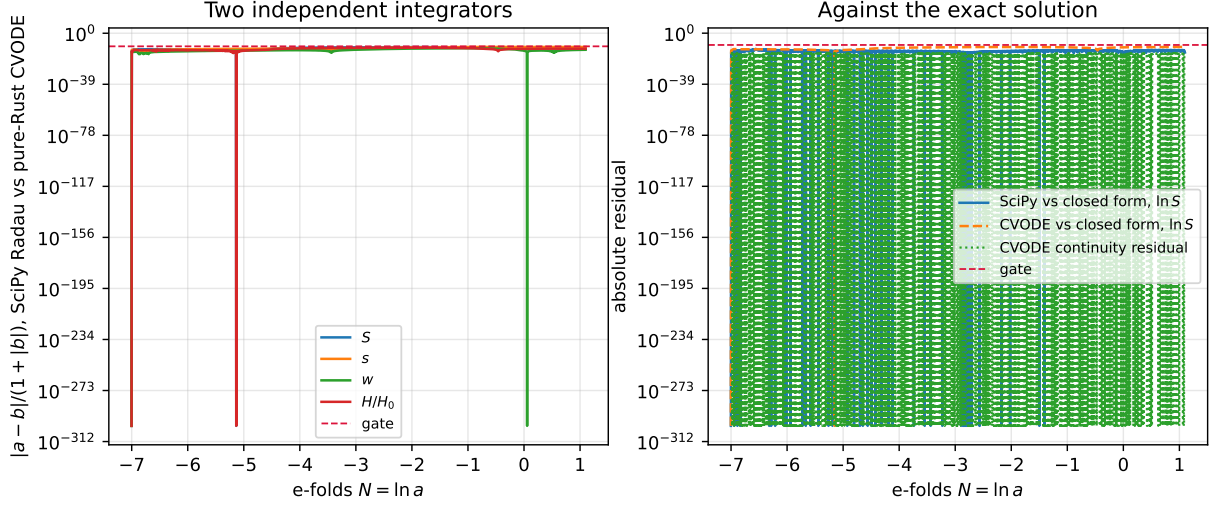


Figure 4: Left: SciPy Radau against the pure-Rust CVODE reference. Right: both against the closed-form solution, and the covariant conservation residual. The dashed lines are the gates.

5 Is there a connection to dark matter and dark energy?

Both, from one field.

At early times S is large, the term MS dominates, and because $w_{\text{dust}} = \nu/3 - 1$ vanishes exactly at $\nu = 3$, the sector is pressureless: it gravitates as *dark matter*. At late times S is small, and since $0 < n < 1$ the term λS^n falls far more slowly and comes to dominate, with $w_{\text{potential}} = n\nu/3 - 1$: it gravitates as *dark energy*. The handover needs no second field and no tuning beyond the choice of n ; the numerical solution confirms the transition is monotonic and never crosses $w = -1$, so the model cannot counterfeit phantom behaviour.

Two mechanisms make w evolve, and they are independent:

1. the *potential* mechanism, a generalisation of quintessence: the ratio $\lambda S^{n-1}/M$ grows as $a^{3(1-n)}$, moving the total w from 0 toward $n - 1$;
2. the *geometric* mechanism supplied by `gpt-5.6_bridge`: the bridge field enters ν and shifts both components' w at fixed n , giving the dark-matter-like component a small negative pressure. Nothing in a pure quintessence can do this, and the effect vanishes exactly when the torsion coupling is switched off.

Finally, the benchmark. Switching the torsion coupling off makes $w_{\text{potential}}$ exactly $n - 1$ — a flat, non-evolving equation of state — so that $n = 0.236$ reproduces $w = -0.764$ with *zero* numerical error, while w_{dust} is exactly zero. The Unite-only constant- w benchmark is thus a clean limit of the model rather than a fitted output.

6 What is not claimed

- Only the *background* is computed. No perturbations are evolved, so nothing here constrains structure formation or the microwave background.
- No observational dataset is shipped or used. The value -0.764 is treated as a benchmark to be reproduced as a limit, never as data being fitted; no likelihood is evaluated anywhere.

- The flow law for the bridge field is a phenomenological closure chosen for smoothness, boundedness and analytic integrability. It is not derived from an action.
- `gpt-5.6f_rame` is a pseudo-orthogonal frame gauge transformation. By itself it changes no observable; the physical content lies in the connection.
- The axial-torsion criterion *failed*, and the superiority claim rests instead on containment of both prior geometries as exact endpoints, on the gradient term of (13), and on irremovability.
- Agreement among three solvers establishes numerical correctness, not physical correctness.
- No claim is made that a new fundamental particle has been detected.